

- This exam contains 14 pages (including the front and back of this cover page) and 16 problems. Check to see if any pages are missing.
- This is the third exam and will count for 20% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 90 points total in this exam, as well as 1 bonus questions worth a total of 3 possible points.

Page:	3	4	5	6	8	9	10	11	12	13	Total
Points:	12	10	10	15	10	5	10	5	5	8	90
Score:											

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

For readability in the key, correct answers are colored in red, e.g. **Correct Answer**, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

2 pts

(a) $\sec(x) \csc(x) \cot(x) = \csc^2(x)$.

☒ True

☐ False

2 pts

(b) $\frac{\sec(x)}{\tan(x)} = \cos(x)$.

☐ True

☒ False

2 pts

(c) $\tan\left(\frac{3\pi}{4}\right) = \frac{-1 + \sqrt{3}}{1 + \sqrt{3}}$

☐ True

☒ False

2 pts

(d) The absolute value of the complex number $-5.6 - 6.8i$ is approximately 8.81.

☒ True

☐ False

2 pts

(e) In polar form, $6i = 6\text{cis}\left(\frac{\pi}{2}\right)$

☒ True

☐ False

2 pts

(f) $\frac{\cot(\theta)}{\tan(\theta)} = \csc^2(\theta) \sec^2(\theta)$

☐ True

☒ False

5 pts

2. Simplify the expression

$$\frac{\tan\left(\frac{4}{7}x\right) + \tan\left(\frac{3}{7}x\right)}{1 - \tan\left(\frac{4}{7}x\right)\tan\left(\frac{3}{7}x\right)}$$

☒ (A) $\tan(x)$

☐ (B) $\tan(7x)$

☐ (C) $\tan(3x)$

☐ (D) $\tan(-4x)$

This is the sum formula for tangent, with $\alpha = \frac{4}{7}x$ and $\beta = \frac{3}{7}x$. So,

$$\frac{\tan\left(\frac{4}{7}x\right) + \tan\left(\frac{3}{7}x\right)}{1 - \tan\left(\frac{4}{7}x\right)\tan\left(\frac{3}{7}x\right)} = \tan\left(\frac{4}{7}x + \frac{3}{7}x\right) = \tan(x).$$

5 pts

3. Find the exact value of $\tan\left(\frac{13\pi}{12}\right)$.

☐ (A) -1

☐ (B) $\frac{1}{2}$

☐ (C) 1

☒ (D) $2 - \sqrt{3}$

Since

$$\frac{13\pi}{12} = \frac{\frac{13\pi}{6}}{2},$$

we use the half-angle formula for tangent. Notice, $\frac{13\pi}{12}$ is in quadrant 3, so the tangent will be positive.

$$\begin{aligned}\tan\left(\frac{13\pi}{12}\right) &= \frac{1 - \cos\left(\frac{13\pi}{6}\right)}{\sin\left(\frac{13\pi}{6}\right)} \\ &= \frac{1 - \frac{\sqrt{3}}{2}}{\frac{1}{2}} \\ &= 2 - \sqrt{3}.\end{aligned}$$

5 pts

4. Choose the option below that is equal to
- $4 \sin(7x) \cos(7x)$
- .

☒ (A) $2 \sin(14x)$

☐ (B) $2 \cos(14x)$

☐ (C) $\sin\left(\frac{7}{2}x\right)$

☐ (D) $\sin(3x) - \sin(4x)$

Recall that the double angle formula says

$$\sin(2\theta) = 2 \sin(\theta) \cos(\theta).$$

So, setting $\theta = 7x$, we have

$$2 \sin(7x) \cos(7x) = \sin(2(7x)) = 2 \sin(14x)$$

5 pts

5. Choose the option below that is equal to
- $10 \sin(9x) \sin(5x)$
- .

☐ (A) $5 [\sin(14x) - \cos(4x)]$

☒ (B) $5 [\cos(4x) - \cos(14x)]$

☐ (C) $10 \cos(2x) \cos(7x)$

☐ (D) $20 \sin(7x) \cos(2x)$

The product to sum formula says

$$\sin(\alpha) \sin(\beta) = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)].$$

So, setting $\alpha = 9x$ and $\beta = 5x$, we have

$$\begin{aligned} 10 \sin(9x) \sin(5x) &= \frac{10}{2} [\cos(9x - 5x) - \cos(9x + 5x)] \\ &= 5 [\cos(4x) - \cos(14x)] \end{aligned}$$

5 pts

6. Simplify the expression to one term:

$$\sin(x) \cos(10x) - \cos(x) \sin(10x).$$

☐ (A) $\tan\left(\frac{x}{10}\right)$

☐ (B) $\cos(11x)$

☒ (C) $-\sin(9x)$

☐ (D) $\cos(9x)$

Using the one of the sum/difference formulas for sine, we have

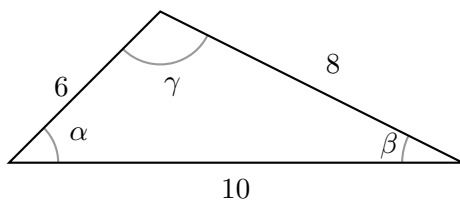
$$\sin(\alpha - \beta) = \sin(\alpha) \cos(\beta) - \cos(\alpha) \sin(\beta).$$

So, setting $\alpha = x$ and $\beta = 10x$, we have

$$\begin{aligned}\sin(x) \cos(10x) - \cos(x) \sin(10x) &= \sin(x - 10x) \\ &= \sin(-9x) \\ &= -\sin(9x)\end{aligned}$$

10 pts

7. Find the area of the triangle below. Round your answer to the nearest tenth.



☐ (A) 31.9

☐ (B) 281.3

☐ (C) 303.6

☒ (D) 24

Using Heron's formula, we have

$$s = \frac{6 + 8 + 10}{2} = 12, \quad A = \sqrt{12(12 - 6)(12 - 8)(12 - 10)} \approx 24$$

8. Solve the equation exactly on the interval $0 \leq \theta < 2\pi$.

$$\frac{\sin(2\theta)}{\sqrt{3}\sin(\theta)} = 1$$

- Ⓐ $\theta = \frac{\pi}{12}, \theta = \frac{11\pi}{12}, \theta = \frac{13\pi}{12},$ and $\theta = \frac{23\pi}{12}$
 Ⓑ $\theta = \frac{\pi}{6}, \theta = \frac{11\pi}{6}, \theta = \frac{13\pi}{6},$ and $\theta = \frac{23\pi}{6}$
 Ⓒ $\theta = \frac{\pi}{24}, \theta = \frac{11\pi}{24}, \theta = \frac{13\pi}{24},$ and $\theta = \frac{23\pi}{24}$
 Ⓓ $\theta = \frac{\pi}{3}, \theta = \frac{11\pi}{3}, \theta = \frac{13\pi}{3},$ and $\theta = \frac{23\pi}{3}$

NOTE: There was a small error in the statement of this problem. Namely, the correct answer had solutions belonging to the interval $0 \leq \theta < 4\pi$ instead of $0 \leq \theta < 2\pi$. This shouldn't have effected your answer, provided you were actually attempting to *do* the problem. If you were just starting with the multiple choice answers and working backwards, this definitely did cause a problem. In light of this, the question was removed from the scoring of the exam and the total number of points on the exam was reduced accordingly.

First, recall the double angle formula

$$\sin(2\theta) = 2\sin(\theta)\cos(\theta).$$

Using this, we then can see that

$$\frac{\sin(2\theta)}{\sin(\theta)} = 2\cos(\theta).$$

So, we have

$$\begin{aligned}\frac{\sin(2\theta)}{\sqrt{3}\sin(\theta)} &= 1 \\ \frac{\sin(2\theta)}{\sin(\theta)} &= \sqrt{3} \\ 2\cos(\theta) &= \sqrt{3} \\ \cos(\theta) &= \frac{\sqrt{3}}{2}\end{aligned}$$

Now, we solve $\cos(x) = \frac{\sqrt{3}}{2}$ on $0 \leq x \leq 4\pi$ using the unit circle. This yields

$$x = \frac{\pi}{6}, \quad x = \frac{11\pi}{6}, \quad x = \frac{13\pi}{6}, \quad \text{and} \quad x = \frac{23\pi}{6}.$$

10 pts

9. Solve the equation exactly on the interval $[0, 2\pi)$.

$$\sin^2(x)(1 - \cos^2(x)) - \cos^2(x)(1 - \sin^2(x)) = 0$$

- (A) $x = \frac{\pi}{12}$, $x = \frac{\pi}{4}$, $x = \frac{5\pi}{12}$, and $x = \frac{7\pi}{12}$.
(B) $x = \frac{\pi}{8}$, $x = \frac{3\pi}{8}$, $x = \frac{5\pi}{8}$, and $x = \frac{7\pi}{8}$.
(C) $x = \frac{\pi}{4}$, $x = \frac{3\pi}{4}$, $x = \frac{5\pi}{4}$, and $x = \frac{7\pi}{4}$.
(D) $x = \frac{\pi}{2}$, $x = \frac{3\pi}{2}$, $x = \frac{5\pi}{2}$, and $x = \frac{7\pi}{2}$.

First, recall the double angle formula $\sin^2(x) + \cos^2(x) = 1$. Using this, we then can see that $\cos^2(x) = 1 - \sin^2(x)$ and $\sin^2(x) = 1 - \cos^2(x)$. So, we have

$$\begin{aligned}\sin^2(x)(1 - \cos^2(x)) + \cos^2(x)(1 - \sin^2(x)) &= 0 \\ \sin^2(x)\sin^2(x) - \cos^2(x)\cos^2(x) &= 0 \\ \sin^4(x) - \cos^4(x) &= 0 \\ (\sin^2(x) - \cos^2(x))(\sin^2(x) + \cos^2(x)) &= 0 \\ \sin^2(x) - \cos^2(x) &= 0.\end{aligned}$$

Now, recall the double angle formula gives that

$$\cos(2x) = \cos^2(x) - \sin^2(x).$$

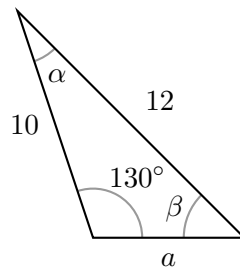
So, solving $\sin^2(x) - \cos^2(x) = 0$ is the same as solving $\cos(2x) = 0$. Set $2x = \theta$ and solve $\cos(\theta) = 0$ on $0 \leq x \leq 4\pi$ using the unit circle. This yields

$$\theta = \frac{\pi}{2}, \quad \theta = \frac{3\pi}{2}, \quad \theta = \frac{5\pi}{2}, \quad \text{and} \quad \theta = \frac{7\pi}{2}.$$

Since $\theta = 2x$, it follows that $x = \frac{\theta}{2}$, we have our final answer:

$$x = \frac{\pi}{4}, \quad x = \frac{3\pi}{4}, \quad x = \frac{5\pi}{4}, \quad \text{and} \quad x = \frac{7\pi}{4}.$$

5 pts

10. Solve for a in the triangle below. Round your answer to the nearest tenth.

Ⓐ 2.2

Ⓑ 4.1

Ⓒ 2.8

Ⓓ 3.1

Using the law of sines:

$$\frac{\sin(\beta)}{10} = \frac{\sin(130^\circ)}{12}$$

$$\sin(\beta) = \frac{10}{12} \sin(130^\circ)$$

$$\beta = \sin^{-1}\left(\frac{10}{12} \sin(130^\circ)\right) \approx 39.67^\circ$$

So, $\alpha \approx 180^\circ - 130^\circ - 39.67^\circ = 10.33^\circ$. Using the law of sines again, this gives

$$\frac{\sin(10.33^\circ)}{a} \approx \frac{\sin(130^\circ)}{12}$$

$$a \approx \frac{12 \sin(10.33^\circ)}{\sin(130^\circ)} \approx 2.8.$$

5 pts

11. Convert the point $(2\sqrt{2}, 2\sqrt{2})$ from rectangular coordinates into polar coordinates (r, θ) with $0 \leq r$ and $0 \leq \theta < 2\pi$.

Ⓐ $\left(2\sqrt{2}, \frac{3\pi}{4}\right)$

Ⓑ $\left(8, \frac{\pi}{4}\right)$

Ⓒ $\left(4, \frac{\pi}{4}\right)$

Ⓓ $\left(4, \frac{5\pi}{4}\right)$

First, we find the radius r :

$$r^2 = x^2 + y^2$$

$$r^2 = (2\sqrt{2})^2 + (2\sqrt{2})^2$$

$$r^2 = 16$$

$$r = 4.$$

Now, we use the unit circle to notice that

$$\cos(\theta) = \frac{2\sqrt{2}}{4} = \frac{\sqrt{2}}{2}, \quad \sin(\theta) = \frac{2\sqrt{2}}{4} = \frac{\sqrt{2}}{2}.$$

So, it's easy to see that $\theta = \frac{\pi}{4}$.

5 pts

12. Let z_1 and z_2 be complex numbers given in polar form by

$$z_1 = 25\text{cis}\left(\frac{13\pi}{12}\right), \text{ and } z_2 = 5\text{cis}\left(\frac{\pi}{12}\right).$$

Find $\frac{z_1}{z_2}$ in polar form.

Ⓐ $5\text{cis}\left(\frac{3\pi}{4}\right)$

Ⓑ $\sqrt{5}\text{cis}\left(\frac{3\pi}{4}\right)$

Ⓒ $5\text{cis}\left(\frac{7\pi}{6}\right)$

Ⓓ $\frac{1}{5}\text{cis}\left(\frac{3\pi}{4}\right)$

NOTE: There was a small error in the statement of this problem. Namely, the problem—as I had intended to state it—should have given z_2 as $z_2 = 5\text{cis}\left(\frac{\pi}{3}\right)$ instead of $z_2 = 5\text{cis}\left(\frac{\pi}{12}\right)$. This error was pointed out during the exam (thanks to the diligent attention to detail of one of your classmates) and the question was withdrawn via an announcement made during the exam. Since the error was discovered during the course of the exam, all students received full credit for this problem.

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\text{cis}(\theta_1 - \theta_2) = \frac{25}{5}\text{cis}\left(\frac{13\pi}{12} - \frac{\pi}{3}\right) = 5\text{cis}\left(\frac{3\pi}{4}\right).$$

5 pts

13. Solve the system by any method.

$$\begin{cases} 5x - 5y = 5/4 \\ 4x - 5y = 1 \end{cases}$$

$$x = \underline{\frac{1}{4}}, \quad y = \underline{0}$$

We solve the system for x and y using whatever method seems easiest. Here, we opt for elimination to solve for x :

$$\begin{array}{r} 5x - 5y = \frac{5}{4} \\ - \quad (4x - 5y = 1) \\ \hline (5 - 4)x + (-5 + 5)y = \frac{5}{4} - 1 \\ x = \frac{1}{4} \end{array}$$

Now, we use the value of x we found to solve for y by substituting it in to one of the original equations in our system. We choose the first equation, $5x - 5y = \frac{5}{4}$:

$$\begin{array}{r} 5 \left(\frac{1}{4} \right) - 5y = \frac{5}{4} \\ \frac{5}{4} - 5y = \frac{5}{4} \\ -5y = 0 \\ y = 0 \end{array}$$

So, we have our final answer: $x = \frac{1}{4}$ and $y = 0$.

5 pts

14. The admission fee at a small fair is \$1.50 for children and \$4.00 for adults. On a certain day, 2200 people enter the fair and \$5050 is collected. How many children and how many adults attended?

Adults: 700,Children: 1500

Let x denote the number of adults and let y denote the number of children. Then, we have the system

$$\begin{cases} x + y &= 2200 \\ 4x + 1.5y &= 5050 \end{cases}$$

So, we solve the system for x and y using whatever method seems easiest. Here, we opt for substitution and solve the first equation for x :

$$x + y = 2200$$

$$x = 2200 - y$$

Now, we substitute our formula for x in to the second equation

$$4x + 1.5y = 5050$$

$$4(2200 - y) + 1.5y = 5050$$

$$8800 - 2.5y = 5050$$

$$y = \frac{5050 - 8800}{-2.5} = 1500.$$

Using our equation from the first step, this gives $x = 2200 - 1500 = 700$. So, we have our answer: 700 adults and 1500 children attended the fair.

8 pts

15. Solve the system by any method.

$$\begin{cases} x - 2y + 3z &= 7 \\ 2x + y + z &= 4 \\ -3x + 2y - 2z &= -10 \end{cases}$$

$$x = \underline{2}$$

$$y = \underline{-1}$$

$$z = \underline{1}$$

First, we use elimination remove a variable and solve for one of the remaining variables in terms of the other. Here, we opt to use the first equation and two times the second equation to solve for x in terms of z :

$$\begin{array}{r} x - 2y + 3z = 7 \\ + \quad (4x + 2y + 2z = 8) \\ \hline (1 + 4)x + (-2 + 2)y + (3 + 2)z = 7 + 8 \\ 5x + 5z = 15 \\ x + z = 3 \\ x = 3 - z. \end{array}$$

Now, we add the first equation and the third equation to eliminate y and solve for z by substituting our equation for x in terms of z :

$$\begin{array}{r} x - 2y + 3z = 7 \\ + \quad (-3x + 2y - 2z = -10) \\ \hline (1 - 3)x + (-2 + 2)y + (3 - 2)z = 7 - 10 \\ -2x + z = -3 \\ -2(3 - z) + z = -3 \\ -6 + 2z + z = -3 \\ 3z = 3 \\ z = 1 \end{array}$$

Thus, $z = 1$. Using our formula for x , we get that $x = 3 - 1 = 2$. We substitute these values in to any of the equations above to solve for y . We choose the second equation:

$$\begin{array}{r} 2x + y + z = 4 \\ 2(2) + y + 1 = 4 \\ y = -1. \end{array}$$

So, we have our answer: $x = 2$, $y = -1$, and $z = 1$

3 (bonus) 16. Solve the system by any method.

$$\begin{cases} y &= x^2 - 2x + 8 \\ y &= 5x - 2 \end{cases}$$

Substitute $y = 5x - 2$ into the first equation, and then solve for x :

$$\begin{aligned} y &= x^2 - 2x + 8 \\ 5x - 2 &= x^2 - 2x + 8 \\ 0 &= x^2 - 7x + 10 \\ 0 &= (x - 2)(x - 5). \end{aligned}$$

So, $x = 2$ and $x = 5$. We substitute both values of x in to the second equation $y = 5x - 2$ to find y .

$$\begin{aligned} y &= 5(2) - 2 = 8 \\ y &= 5(5) - 2 = 23. \end{aligned}$$

So, we have our answers: $(x, y) = (2, 8)$ and $(x, y) = (5, 23)$.